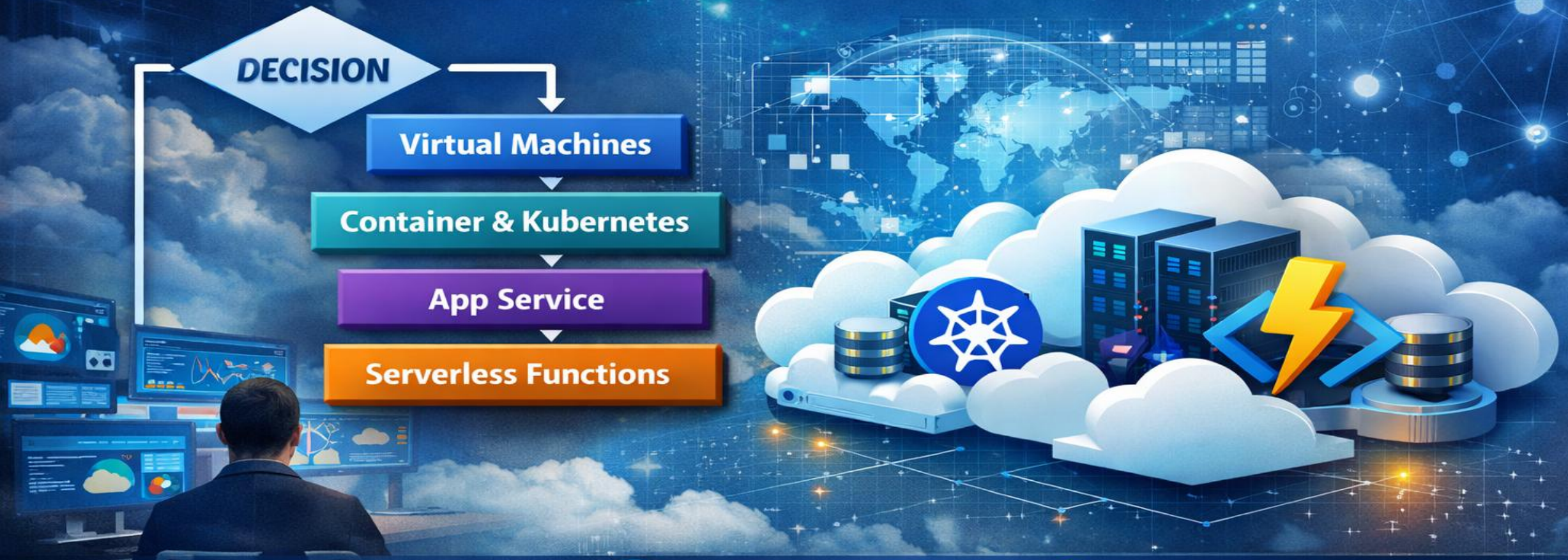
Compute in the Cloud:

The Azure Compute Decision Framework

Choosing the Right Compute Platform in Azure



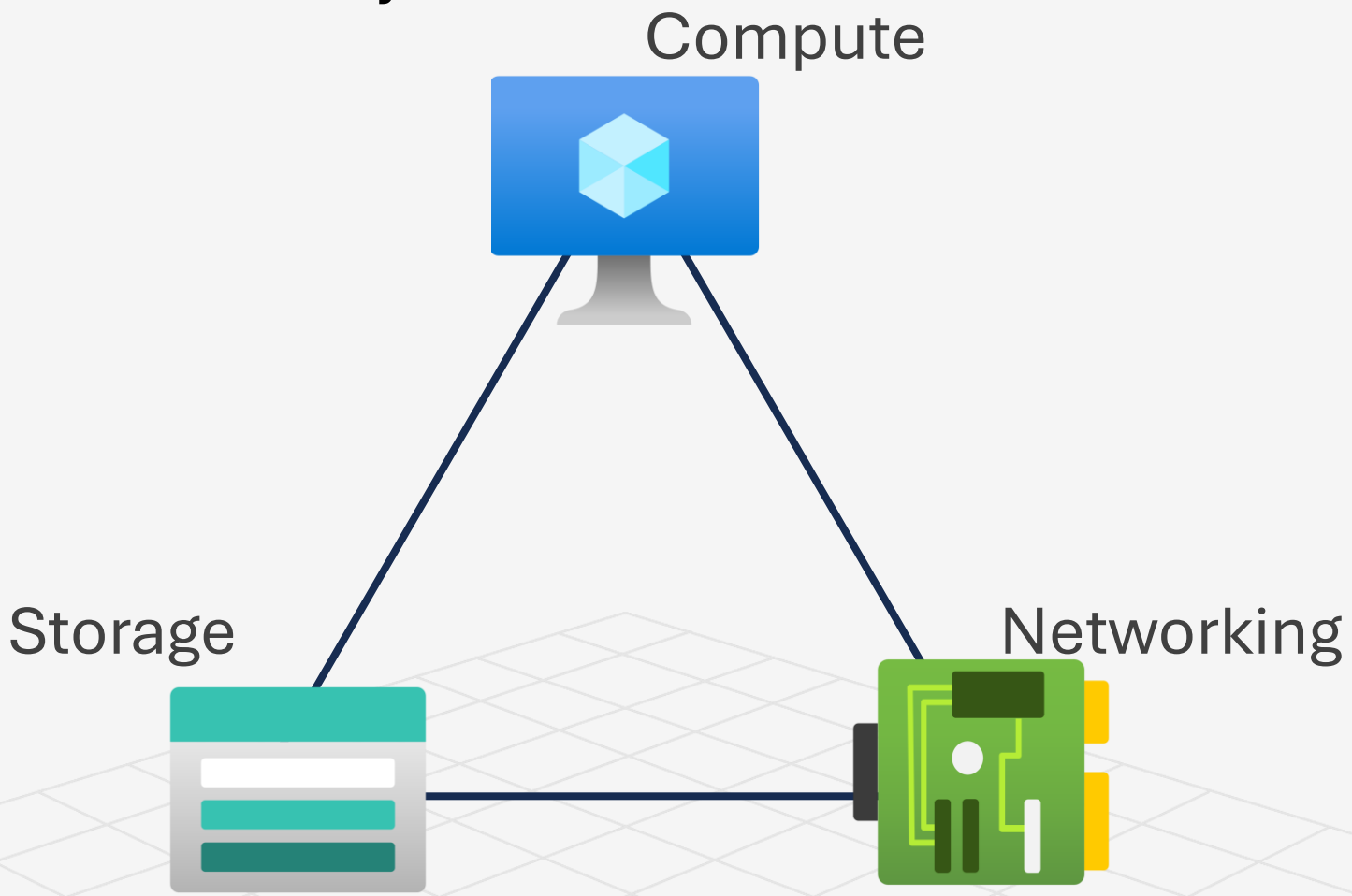
IaaS vs Containers | *PaaS vs Serverless* | *Cost • Scaling • Architecture*

Agenda

What is Azure Compute Infrastructure
The compute spectrum
IaaS vs PaaS vs Serverless
The Azure Compute Decision Framework
Virtual machines
Containers and Kubernetes
Serverless compute
Modern application architectures
Cost and scaling considerations
Real-world architecture examples

What is Compute in the Cloud

Core cloud infrastructure layers



Azure Compute

Collection of cloud-based services in **Microsoft Azure** that provide the infrastructure, tools, and resources for running applications and workloads

Benefits of Azure Compute

Scalability

Automatically scale resources based on demand

Cost Efficiency

Pay-as-you-go pricing and support for reservations

High Availability

Built-in redundancy and geographic distribution options

Security

Compliance with global standards and built-in security measures.

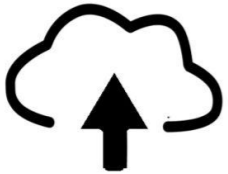
The Azure Compute Ecosystem

Major compute services include:

- Azure Virtual Machines
- Azure Virtual Machine Scale Sets
- Azure Container Instance
- Azure Container Apps
- Azure Kubernetes Service
- Azure Function Apps
- Azure Logic Apps
- Azure App Service

The Compute Spectrum

IaaS
Infrastructure as a Service



Caching
Legacy
Networking
Security
File

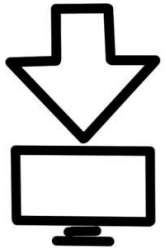
Migrate to It

PaaS
Platform as a Service

Application Development
Decision Support
Web
Streaming

Built on It

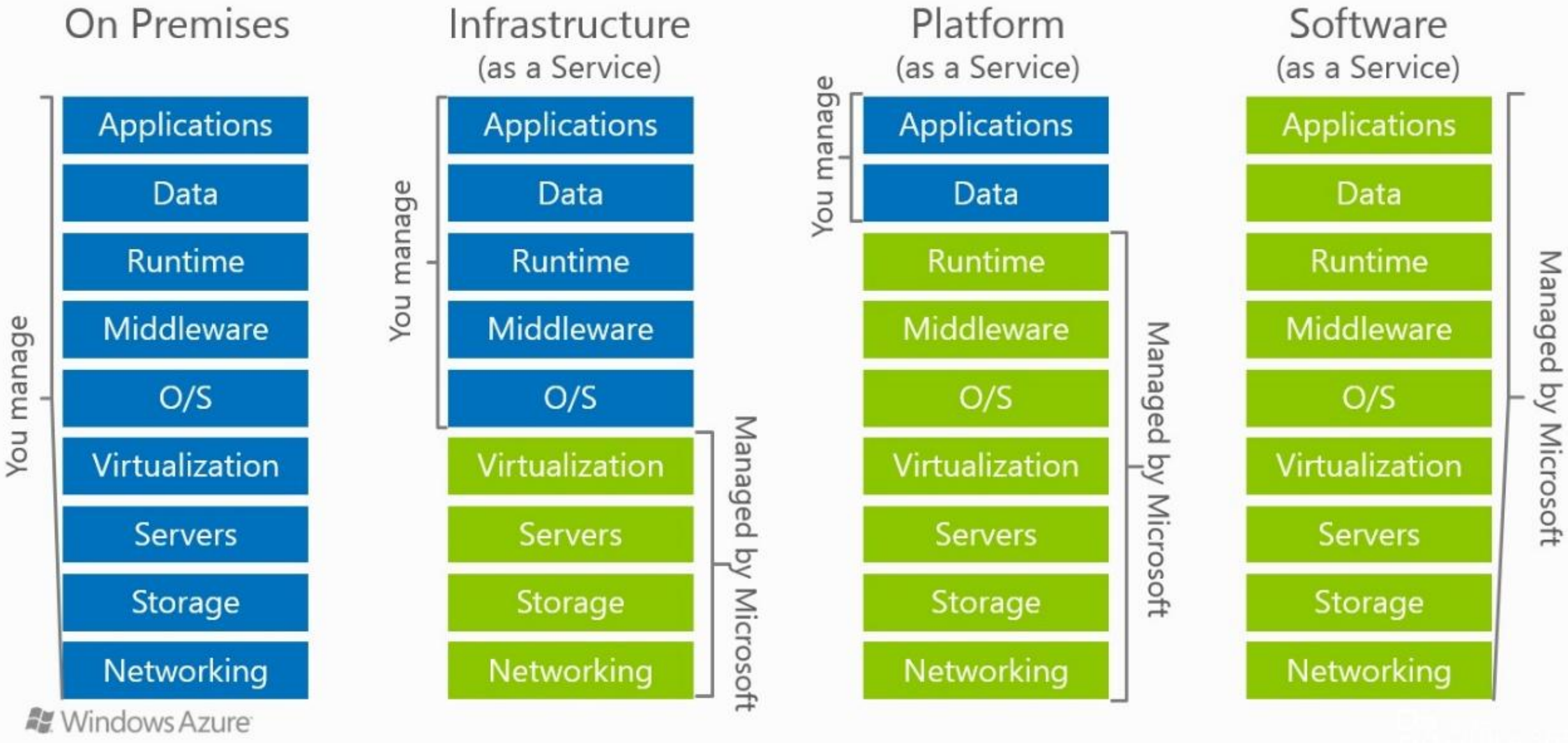
SaaS
Software as a Service



Email
CRM
Collaboration
ERP

Consume

Cloud Models



The Compute Spectrum

Service	Category	Control Level	Use Case
Azure Virtual Machines (VMs)	IaaS	🔴 Full control	Lift & shift, custom OS/apps
VM Scale Sets (VMSS)	IaaS	🔴 Full control \ + autoscale	Scalable VM workloads
Azure Kubernetes Service (AKS)	PaaS (managed control plane)	🟠 Medium-high	Container orchestration
Azure Container Instances (ACI)	PaaS / Serverless	🟡 Low	Simple, short-lived containers
Azure Container Apps (ACA)	PaaS / Serverless	🟡 Low-medium	Microservices, APIs
Azure App Service	PaaS	🟡 Low-medium	Web apps, APIs
Azure Functions	Serverless (FaaS)	🟢 Very low	Event-driven code
Azure Logic Apps	Serverless (Integration SaaS-like)	🟢 Very low	Workflows, automation

Architect Decision Questions

Architects ask several key questions.

1. Do we need OS-level control?
2. Is the application containerized?
3. Does the workload scale dynamically?
4. Is it event-driven?
5. What are the operational requirements?

The Azure Compute Decision Framework

Decision flow

1. OS control required → Virtual Machines
2. Containerized workload → Container platform
3. Event-driven → Serverless
4. Web applications → PaaS

Virtual Machines

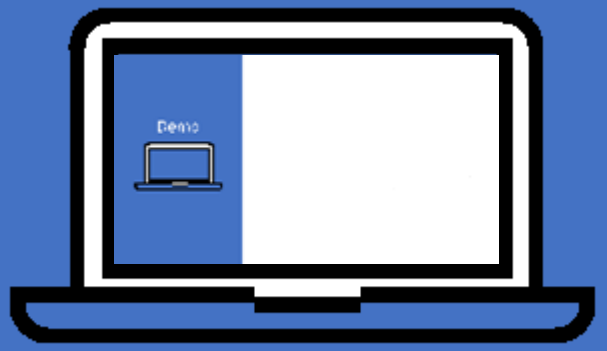
Scalable and customizable virtualized computing resources

Support for multiple operating systems (Windows, Linux)

Use cases:

- Legacy application migration
- Custom workloads
- Development/testing environments

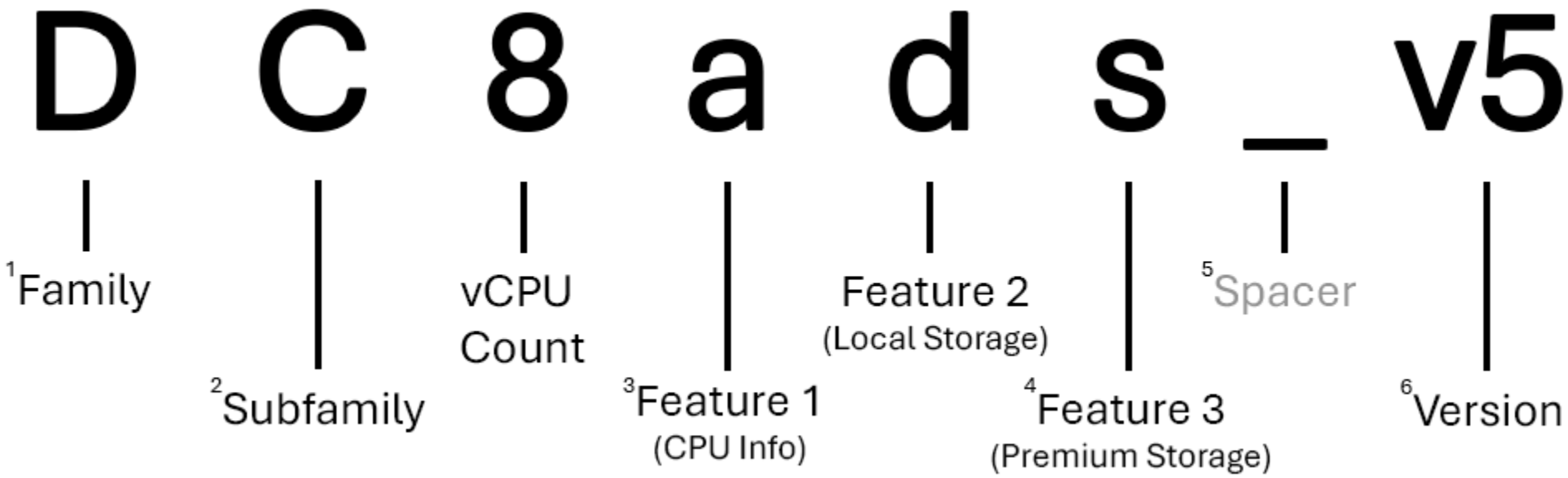
Demo



Virtual Machines

https://learn.microsoft.com/en-us/azure/virtual-machines/

Name structure breakdown



[Family] + *[Sub-family]** + **[# of vCPUs]** + *[Constrained vCPUs]** + **[Additive Features]** + *[Accelerator Type]** + *[Memory Capacity]** + **[Version]**

VM Families

Family	Description
A	Entry-level economical
B	Burstable
D	General Purpose
E,M	Memory Optimized
F	Compute Intensive
L	Storage Optimized
H	High Performance Computing
N	Accelerated (GPU)

VM Subfamilies

Subfamily	Description
B	Memory bandwidth optimized
C	Confidential (for DC, EC, NCC series)
C	Compute intensive (for HC, NC, NCC series)
D	AI training and inference optimized
G	Cloud gaming and remote desktop optimized
V	Visualization and graphics optimized
X	Extra memory

Additive Features

Additive Feature	Description
a	AMD-based processor
b	Remote storage bandwidth optimized
d	Includes local temporary disks
e	Encrypted; contains confidential TDX capabilities
f	Flat ratio of vCPU to memory size
i	Low memory; decreased vCPU to memory ratio
m	Memory intensive
n	Network optimized
o	Increased vCPU to local SSD
p	ARM-based processor
r	RDMA (InfiniBand) secondary network
s	Premium SSD type
t	Tiny memory

Virtual Machine Scale Sets

Automatic scaling of virtual machines.

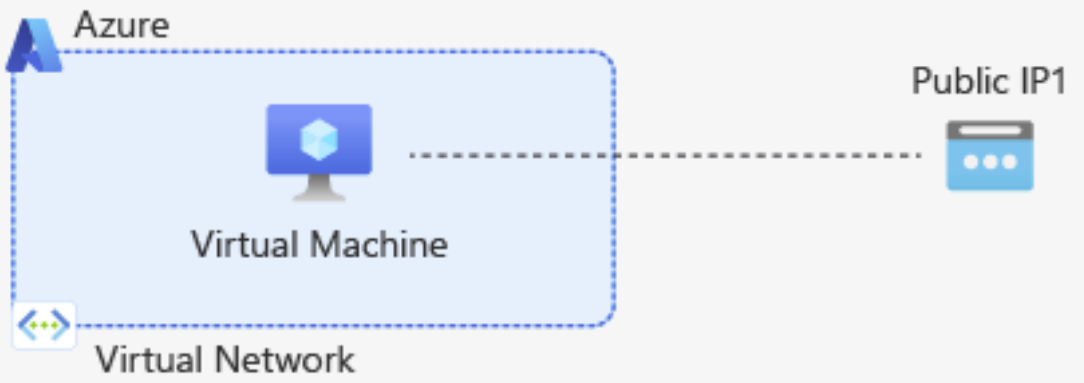
Use cases:

- Hosting large-scale applications
- Scalable backend services for APIs, web apps, or mobile apps
- Big data processing and batch workloads

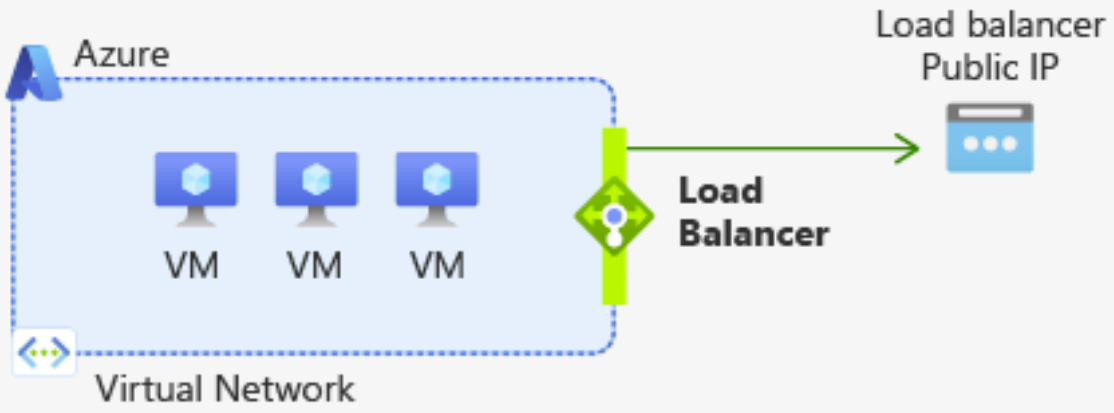
Benefits:

- Elasticity
- High Availability
- Load Balancing

Virtual Machine Scale Sets vs VMs



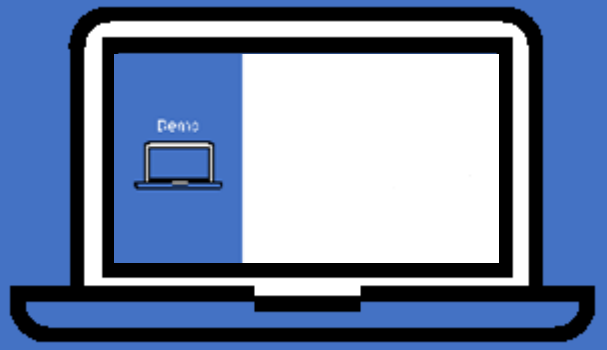
Does not scale as networks grow



Scales as network grow

Image modified from [Default outbound access in Azure - Azure Virtual Network | Microsoft Learn]

Demo



VM Scale Sets

https://learn.microsoft.com/en-us/azure/virtual-machine-scale-sets/

Container Instances

Fast and simple way to run containers without managing virtual machines or requiring a complex orchestration framework

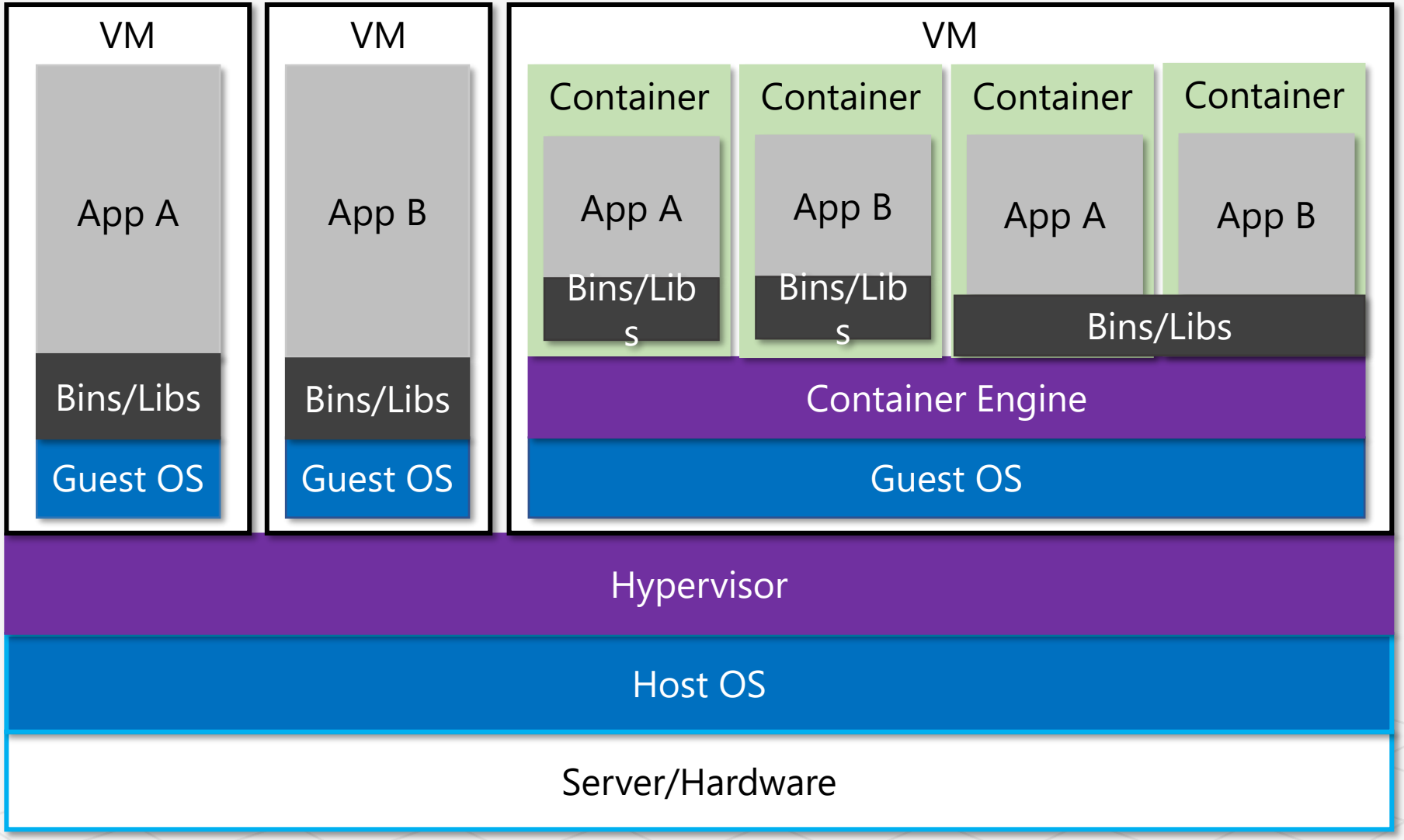
Supports Linux and Windows containers

Per-second billing and automatic scaling of resources

Use Cases:

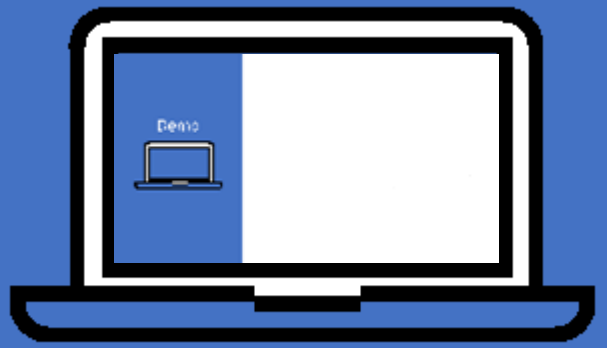
- Development and testing of containerized applications
- Running background jobs or event-driven tasks
- Proof of concept and short-lived workloads

Containers vs VMs



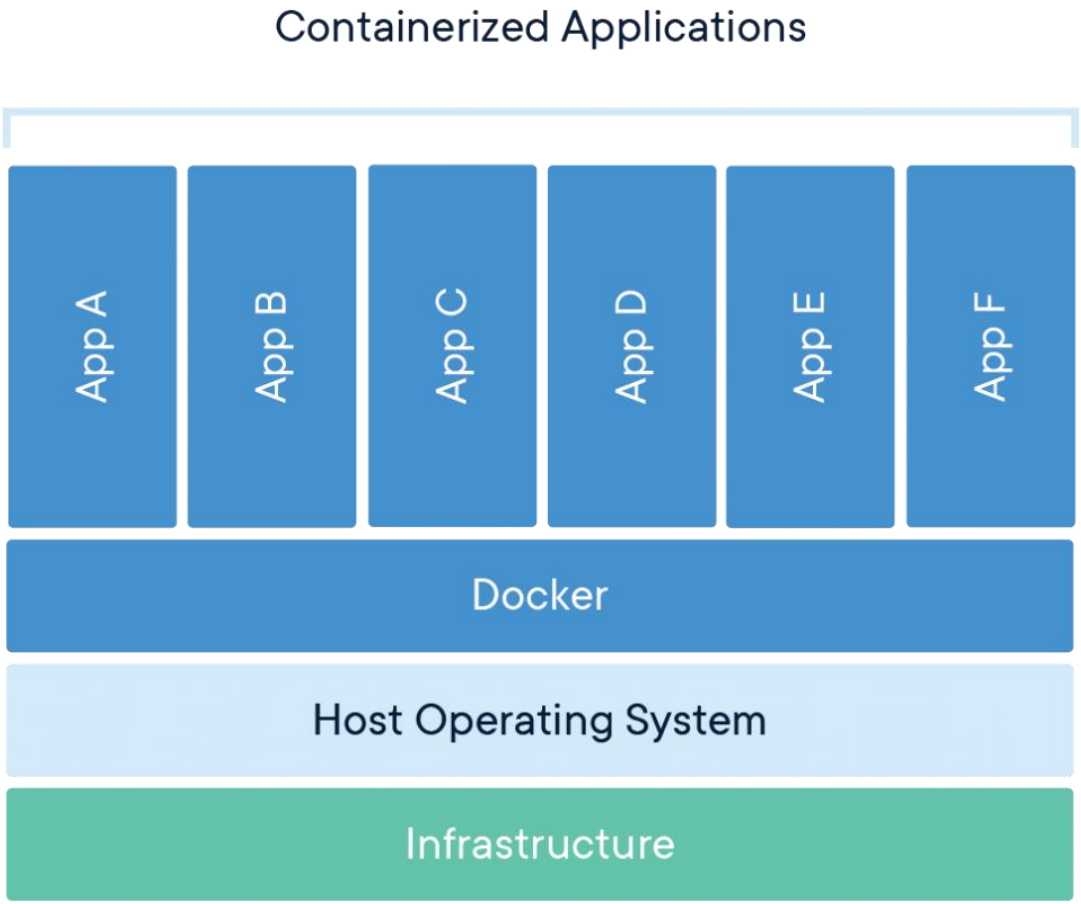
Containers Share OS
And where Appropriate Bins/Libs

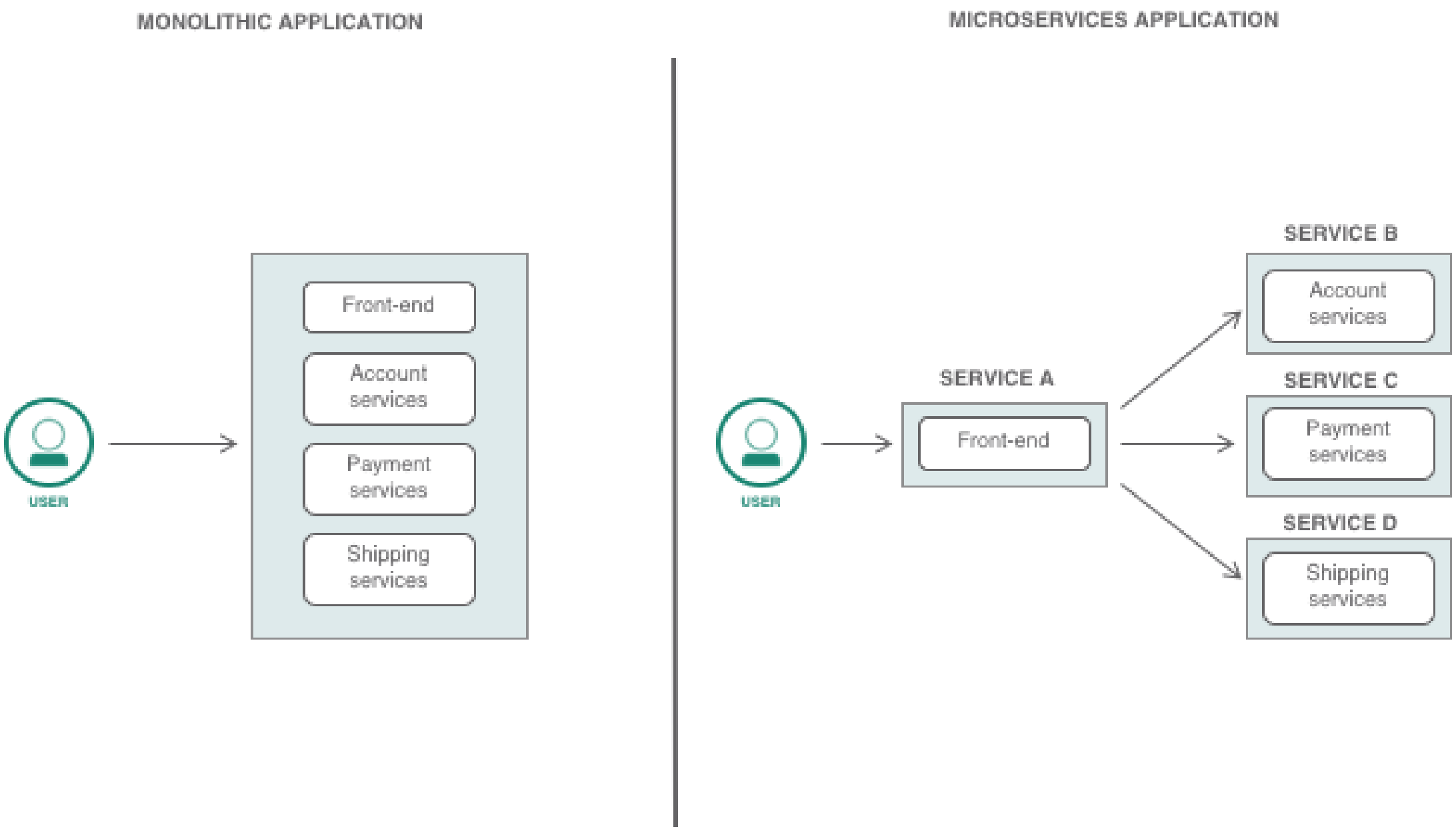
Demo

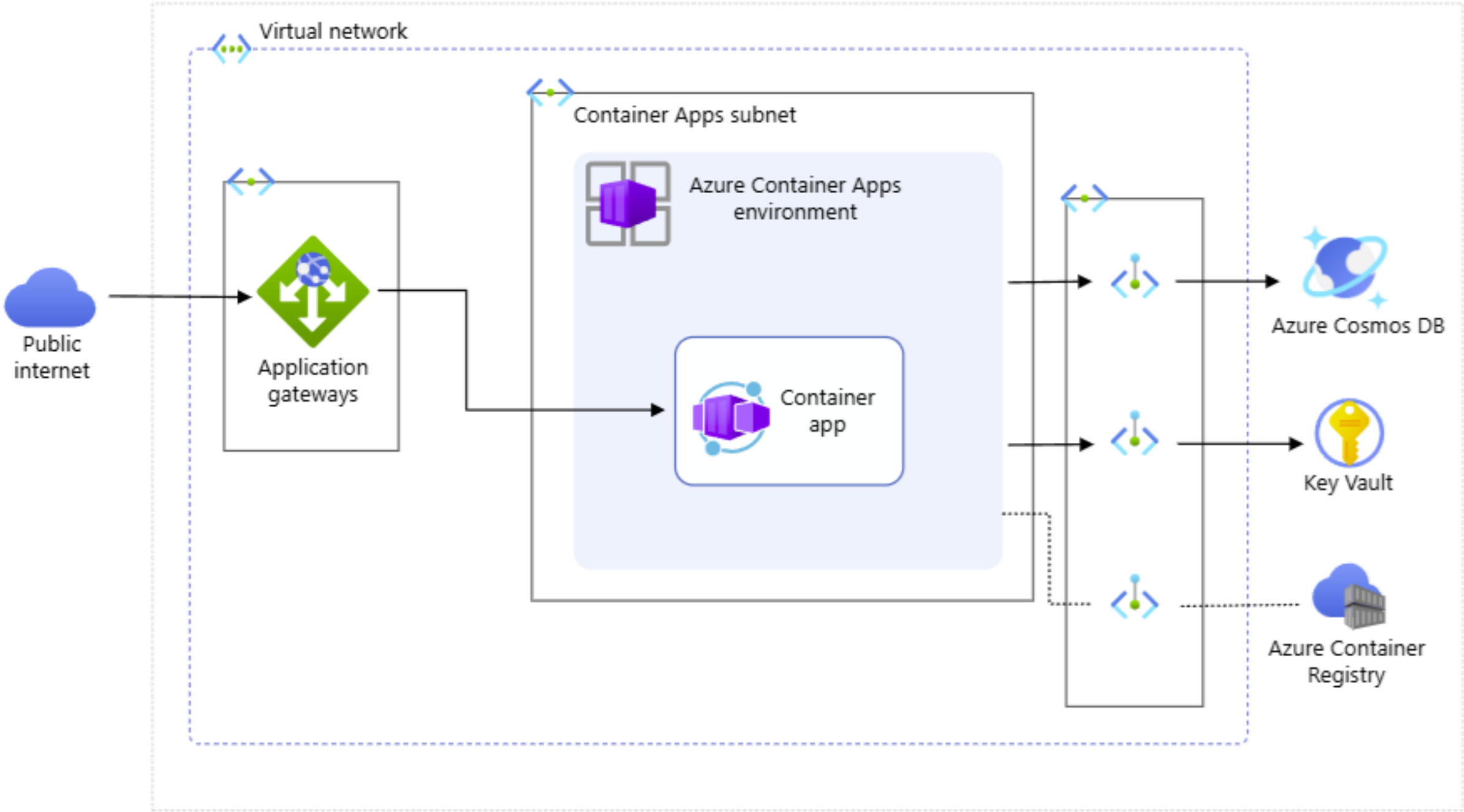


Container Instances

https://learn.microsoft.com/en-us/azure/container-instances/


Containerized Applications
App A
App B
App C
App D
App E
App F
Docker
Host Operating System
Infrastructure


MONOLITHIC APPLICATION
USER
Front-end
Account services
Payment services
Shipping services
MICROSERVICES APPLICATION
USER
SERVICE A
Front-end
SERVICE B
Account services
SERVICE C
Payment services
SERVICE D
Shipping services


Virtual network
Container Apps subnet
Azure Container Apps environment
Public internet
Application gateways
Container app
Azure Cosmos DB
Key Vault
Azure Container Registry

Kubernetes Services

A managed container orchestration service

Simplifies deploying and managing containerized applications using Kubernetes

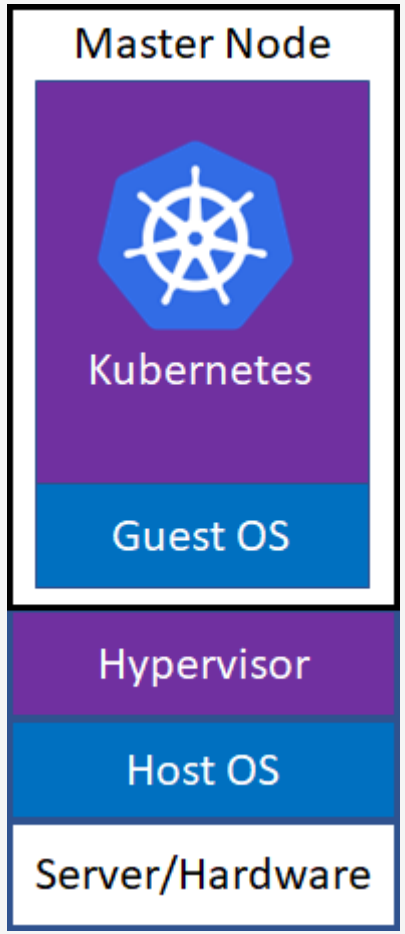
Use cases:

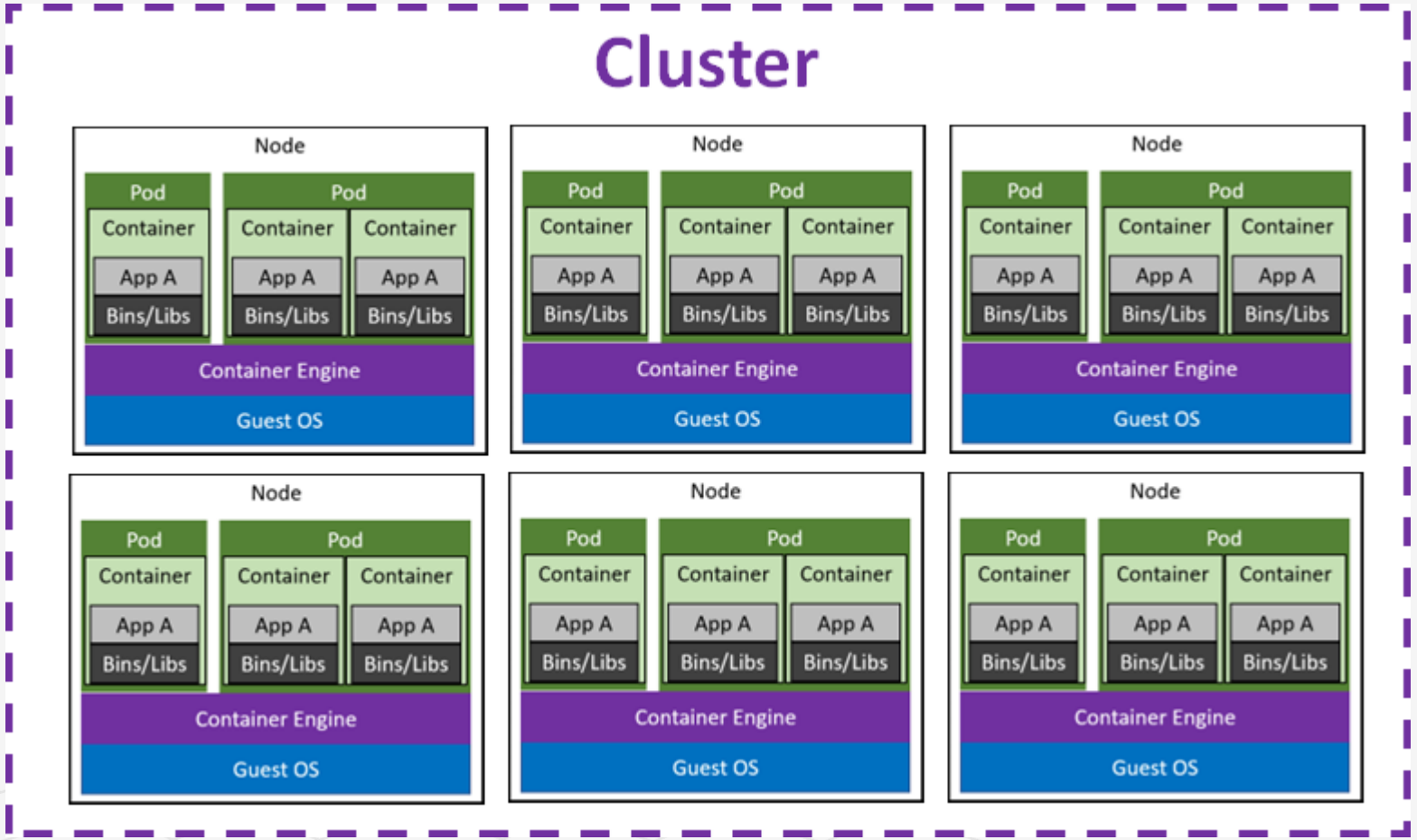
- Microservices architecture
- Containerized application hosting

Benefits:

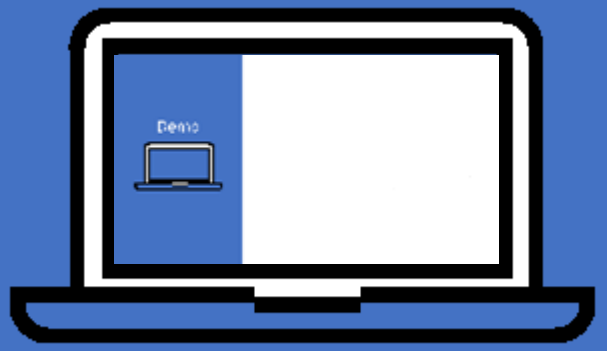
- Scaling
- Self-healing

Kubernetes


Master Node
Kubernetes
Guest OS
Hypervisor
Host OS
Server/Hardware


Cluster
Node
Pod
Container
App A
Bins/Libs
Container Engine
Guest OS

Demo



Kubernetes Services

https://learn.microsoft.com/en-us/azure/aks/

App Services

Managed platform for building, deploying, and scaling web apps, RESTful APIs, and mobile backends

Supports programming languages like .NET, Java, Python, Node.js, and PHP

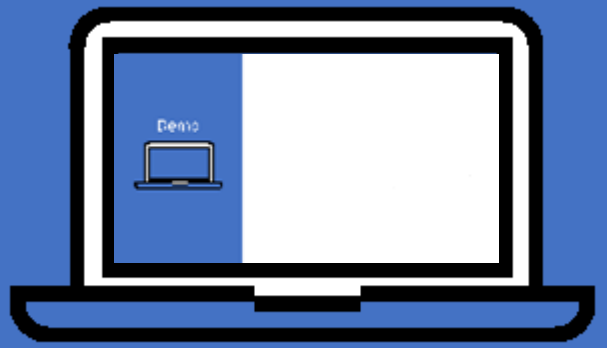
Use cases:

- Web application hosting
- API integration
- Mobile app backend services.

App Services vs VMs

App Services	VMs
PaaS Offering	IaaS Offering
Fully Managed	Full Control

Demo



App Services

https://learn.microsoft.com/en-us/azure/app-service/

Functions

A serverless compute option that runs code on-demand without managing infrastructure

Supports event-driven execution

Use cases:

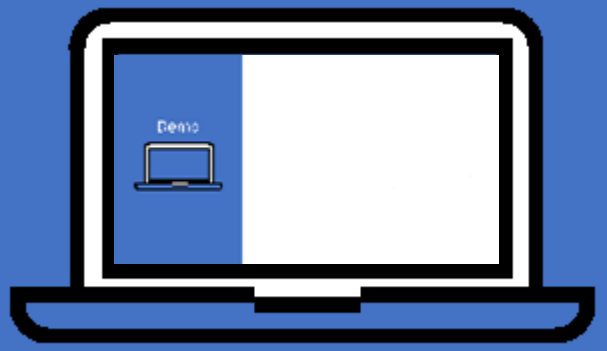
- Event processing
- Automation
- Lightweight application logic

Functions vs App Services

Feature	Azure Functions	Azure App Service
Architecture	Serverless	PaaS
Event-Driven	Yes	No
Use Cases	Lightweight APIs, background tasks, microservices	Web applications
Statefulness	Short-lived tasks	Stateful
Execution Time Limit	Yes	No
Cost	Pay per execution	Fixed higher cost

Table modified from [Azure Functions vs Azure App Service: Choosing the Right Platform for Your Cloud Applications By Robert Dennyson]

Demo



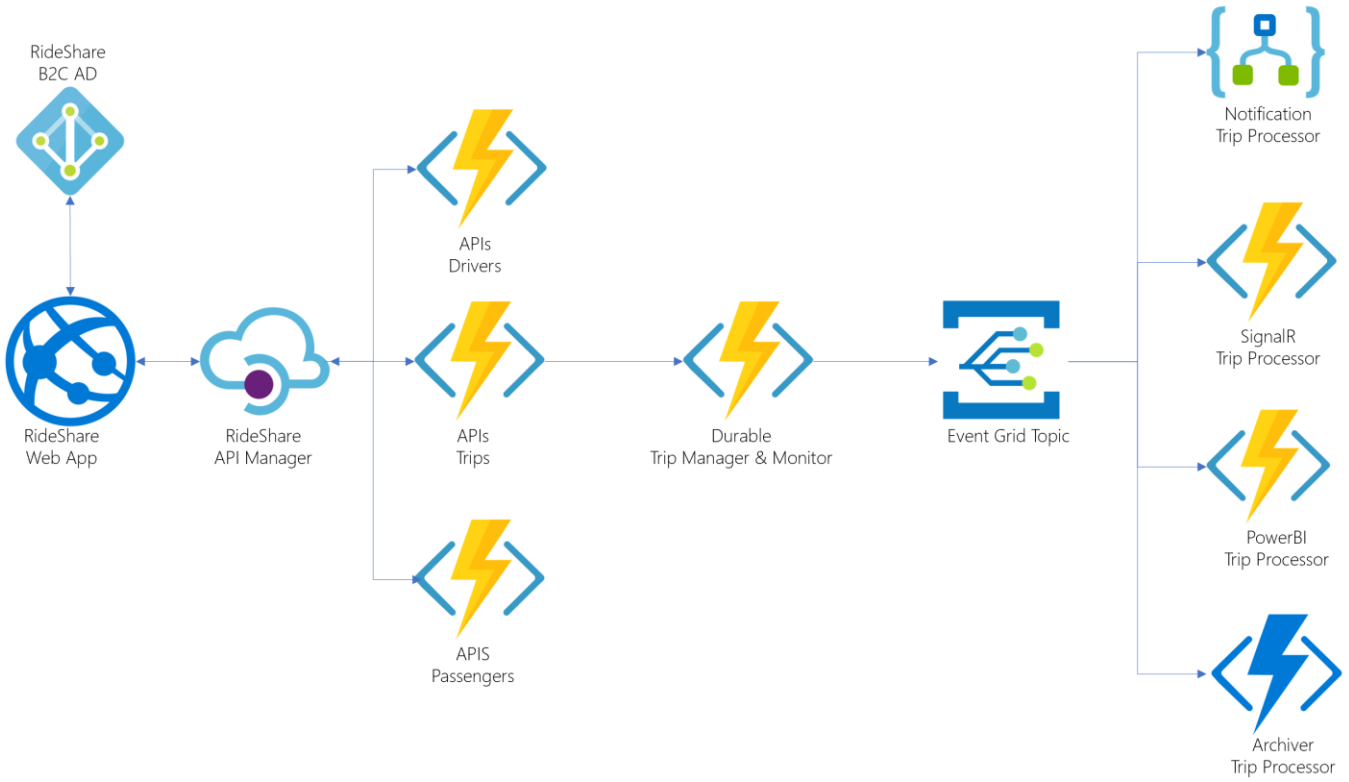
Functions

https://learn.microsoft.com/en-us/azure/azure-functions/

Serverless Compute

Service example

- Azure Functions



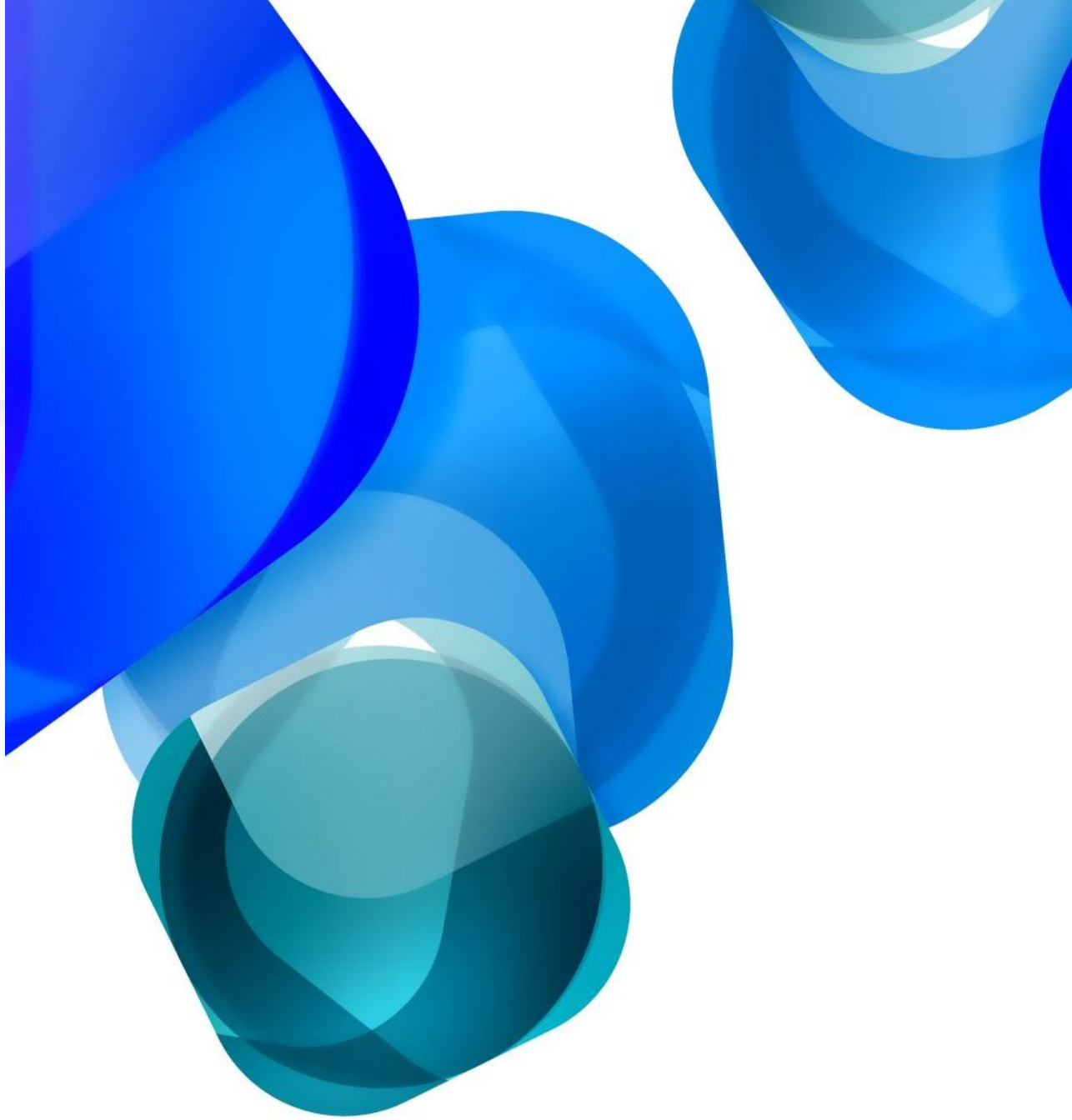
Platform as a Service

Service example

- Azure App Service

Benefits

- built-in scaling
- integrated CI/CD
- managed runtime

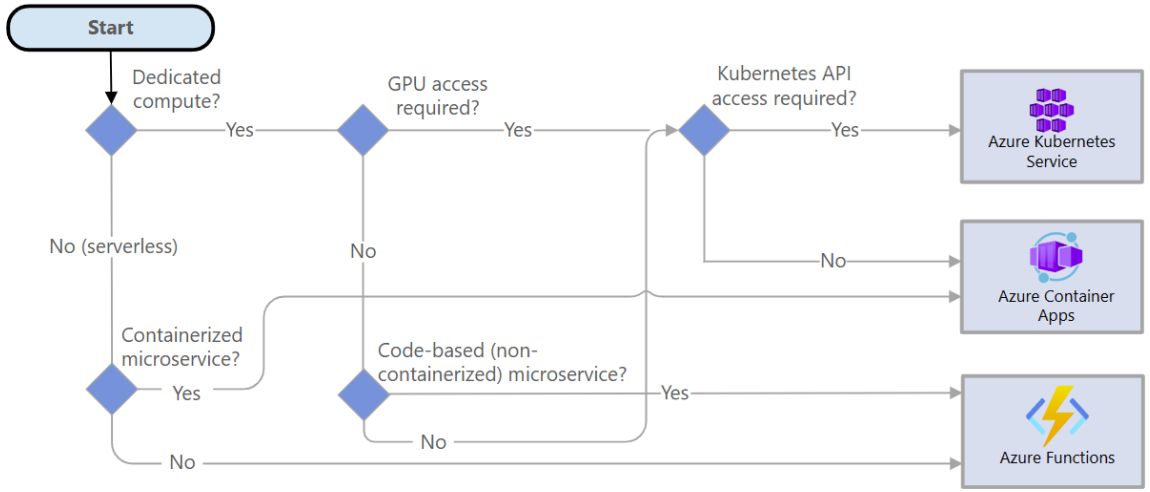


Compute Scaling Models

Scaling approaches

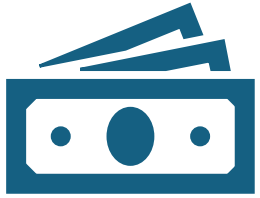
- horizontal scaling
- vertical scaling
- event-driven scaling

Compute option for microservices



- Deploy microservices on dedicated compute platforms, typically by using a microservice orchestrator.
- Deploy microservices on a serverless platform.

https://learn.microsoft.com/en-us/azure/architecture/microservices/design/compute-options



Cost Considerations

Compute cost factors

- runtime duration
- instance size
- scaling behavior
- idle resources

Reliability Architecture

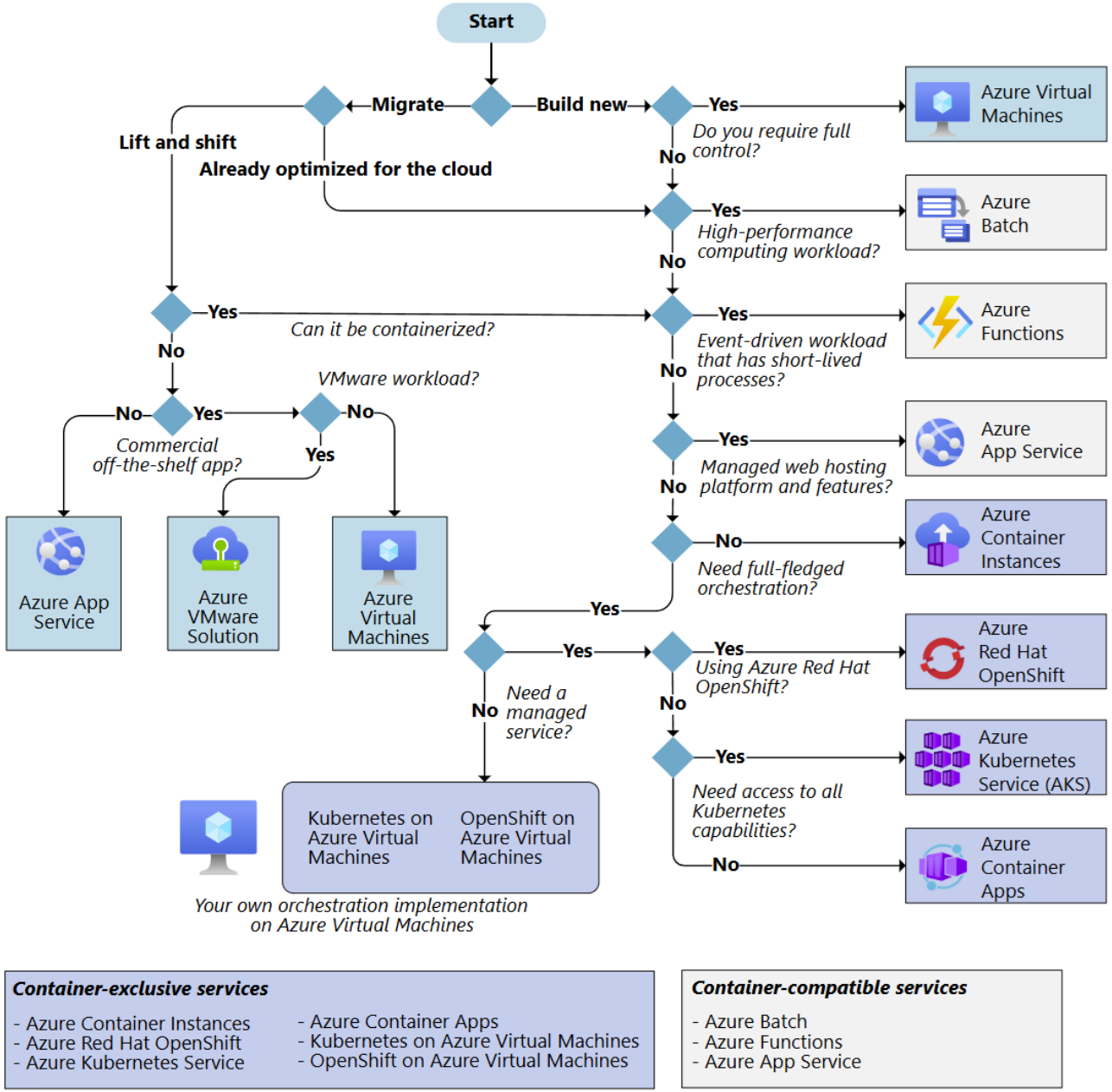
High availability techniques

- Availability zones
- Load balancing
- Redundancy

Decision Framework

Compute choice depends on:

- control
- scalability
- architecture style
- operational complexity



https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree

Summary

Key points

- Azure offers multiple compute platforms
- Architecture decisions affect cost and scalability
- The decision framework simplifies compute selection

Resources

Azure Compute

https://learn.microsoft.com/en-us/azure/?product=compute

AZ-104: Deploy and manage Azure compute resources

https://learn.microsoft.com/en-us/training/paths/az-104-manage-compute-resources/

Link to Slides in GitHub

https://github.com/danielecolon/Azure-Compute-DecisionFramework

Additional Resources

Choose an Azure Compute Service

https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/compute-decision-tree

Choose an Azure Compute Option for Microservices

https://learn.microsoft.com/en-us/azure/architecture/guide/choose-azure-container-service

Choose an Azure Container Service

https://learn.microsoft.com/en-us/azure/architecture/guide/choose-azure-container-service